MATH110C Homework 4

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1 Exercise 53.11.5

Lemma 1.1. Let K/F be an algebraic extension. If $\alpha \in K$ is separable over $F(\alpha^p)$, $\alpha \in F(\alpha^p)$.

Proof. Consider $f(x) = x^p - \alpha^p = (x-\alpha)^p \in F(\alpha^p)[x]$. Notice that $f(\alpha) = 0$. Then, f can't be irreducible in $F(\alpha^p)[x]$ since otherwise α wouldn't be separable as f has multiple roots.

Then, let $f = gh$ for some $g, h \in F(\alpha^p)$ where g is irreducible. Notice that $g = (x-\alpha)^m$, where $1 \leq m < p$. Notice that the last coefficient of g is $-\alpha^m$, which implies that $\alpha^m \in F^{\alpha^p}$.

Then, $\gcd(p, m) = 1$, so there's some $r, s \in \mathbb{Z}$ such that $1 = rp + sm$ by the Bezout Identity. Then, $\alpha = (\alpha^p)^r + (\alpha^m)^s$, so $\alpha \in F(\alpha^p)$. $\square$

Lemma 1.2. Let F with $charF = p > 0$ and K/F be a separable field extension. Then, $K = F(K^p)$, where $K^p := \{x^p : x \in K\}$.

Proof. It suffices to show $K \subset F(K^p)$, the reverse containment is trivial.

Let $\alpha \in K$. Then, α is separable over F, so α is separable over $F(\alpha^p)$. Then, by Lemma 1.1, $\alpha \in F(\alpha^p) \subset K^p$. We thus conclude the proof. $\square$

I couldn't prove the rest of the problem. I wanted to use the Frobenius endomorphism for b, but couldn't work it out.

2 Exercise 53.11.6

Lemma 2.1. Let K/F be an extension of fields. If $\alpha \in K$ is separable over F, $F(\alpha)/F$ is separable.

Proof. Let $p \in F[x]$ be the minimal polynomial of α. Recall that p is separable if and only if

$$\#\{\text{roots of } p \text{ in } L\} = \#\{\tau : F(\alpha) \xrightarrow{\tau} L\}$$

But then,

$$[F(\alpha) : F] = \#\{\text{roots of } p \text{ in } L\} = \#\{\tau : F(\alpha) \xrightarrow{\tau} L\} = [F(\alpha) : F]_s$$

,

so $F(\alpha)/F$ is separable. ☐

Lemma 2.2. If $\alpha_1, \ldots, \alpha_n \in K$ are separable over F, then, $F(\alpha_1, \ldots, \alpha_n)/F$ is separable.

Proof. Notice that if an element $x \in K$ is separable over F, it's separable over any extension of F as well. In particular, α_{i+1} is separable over $F(\alpha_1, \alpha_2, \ldots, \alpha_i)$.

Therefore, $F(\alpha_1, \alpha_2, \ldots, \alpha_i, \alpha_{i+1})/F(\alpha_1, \alpha_2, \ldots, \alpha_i)$ is a separable extension by Lemma 2.1.

A simple inductive argument using K/F is separable if and only if K/E and E/F is separable concludes the proof. ☐

Lemma 2.3. Let $F_{sep} = \{\alpha \in K : \alpha \text{ separable over } F\}$. Then, F_{sep} is a field.

Proof. Let $\alpha, \beta \in F_{sep}$. It suffices to show that $\alpha + \beta, \alpha\beta, \alpha^{-1} \in F_{sep}$.

Since $F(\alpha)/F$ is separable and $\alpha^{-1} \in F(\alpha)$, α^{-1} is separable.

Since $F(\alpha, \beta)/F$ is separable and $\alpha + \beta, \alpha\beta \in F(\alpha, \beta)$, $\alpha + \beta$ and $\alpha\beta$ are also separable.

We thus conclude the proof. ☐

3 Exercise 53.11.9

Lemma 3.1. Let F_0 be a field of positive characteristic p. Let $F = F_0(t_1^p, t_2^p)$ and $L = F_0(t_1, t_2)$. Then, if $\theta \in L \backslash F$, $[F(\theta) : F] = p$.

Proof. Let $\theta \in L \backslash F$. Then, $\theta^p \in F$. Then, θ satisfies $f = x^p - \theta^p = (x - \theta)^p$. f is irreducible since otherwise $\theta \in F$ using the Bezout trick. Thus, $[F(\theta) : F] = p$. $\square$

I couldn't find how to prove that there are infinitely many intermediate fields. We can just keep adjoining the θ's, but I don't see how to formally argue it.

4 Exercise 53.11.10

Lemma 4.1. If $F = F^p$, F is perfect.

Proof. If $F = F^p$, the Frobenius endomorphism is surjective, which implies that F is perfect. $\square$

Lemma 4.2. If F is not perfect, then $\forall r \geq 1 : F \neq F^{p^r}$.

Proof. ? Couldn't prove it. $\square$

5 Exercise 53.11.11

Lemma 5.1. Let K/F be an extension. $\alpha \in K$ is both separable and purely inseparable if and only if $\alpha \in F$.

Proof. $\alpha \in K$ is separable and purely inseparable over F if and only if its irreducible polynomial, $(x-\alpha)^m$, has m distinct roots in some splitting field. Clearly, this happens if and only if $m = 1$. $\square$

6 Exercise 53.11.13

Theorem 6.1. Let K/F be an extension with $char F = p > 0$. Then, the following statements are equivalent:

1. K is purely inseparable over F.
2. The irreducible polynomial of any $\alpha \in K$ is of the form $x^{p^r} - a \in K[x]$.
3. If $\alpha \in K$, then $\alpha^{p^n} \in F$ for some $n \geq 0$.
4. $K_{sep} = F$.
5. K is generated over F by a set of purely inseparable elements.

Proof. Notice that 1 and 5 are equivalent trivially.

Since the only elements that are both separable and purely inseparable over F are elements of F, 4 and 5 are equivalent conditions.

It remains to show that the conditions 1,2,3 are equivalent.

Let's first prove that 1 implies 2.

Assume K is purely inseparable over F. Let $\alpha \in K$ and $f = (x - \alpha)^m$ be the irreducible polynomial of α. Let $m = np^r$ with $\gcd(n, p) = 1$.

Then,

$$(x - \alpha)^m = (x - \alpha)^{np^r} = (x^{p^r} - \alpha^{p^r})^n$$

Then, it suffices to show that $n = 1$. To prove that $n = 1$, it suffices to show that $\alpha^{p^r} \in F$. As f is irreducible, this will imply that $n = 1$. Notice that the coefficient of $x^{p^r(n-1)}$ in f is $\pm n\alpha^{p^r}$, which is in K by definition. Since $\gcd(p, n) = 1$, we have that $\alpha^{p^r} \in F$ and we thus conclude the proof.

Notice that 2 implies 3 is trivial.

Let's now prove that 3 implies 1. Let $\alpha \in K$. We'd like to show that the minimal polynomial f of α in $F[x]$ has the form $(x - \alpha)^m$ for some m. Since $\alpha^{p^n} \in F$ for some $n > 0$, α is a root of $x^{p^n} - \alpha^{p^n} = (x - \alpha)^{pn}$. Since f divides this polynomial, we're done with the proof. $\square$

7 Exercise 53.11.14

Lemma 7.1. Let K/F be an extension with $charF = p > 0$. If $\alpha \in K$ is algebraic over F, α^{p^n} is separable over F for some $n \geq 0$.

Proof. We can assume without loss of generality that α is not separable using Lemma **??**. We induct on the degree of α, which we denote by n. The base case with $n = 1$ is obvious, since $\alpha \in F$. Suppose $n > 1$.

Let f be the minimal polynomial of α^{p^n}. Then, $\exists g \in F[t] : f(t) = g(t^p)$. Notice that the degree of g is less than the degree of f and that $\alpha^{p^{n-1}}$ is a root of g. Thus, g is separable. Then, f is separable. ☐

Lemma 7.2. Let K/F be an algebraic field extension. Then, K/F_{sep} is purely inseparable and $K_{psep} := \{\alpha \in K : \alpha \text{ is purely inseparable over } F\}$ is a field.

Proof. If $charF = 0$, $F_{sep} = K$ and thus K/K is a trivial extension.

Thus, assume $charF = p > 0$. Then, by Lemma 7.1, $\exists n > 0 : \alpha^{p^n} \in F_{sep}$. Then, the third condition from Lemma 6.1 finishes the proof. ☐